

LSSN

Monopartite

$$\eta_{ij} = \alpha - |z_i - z_j| \quad (1)$$

where

- η_{ij} : the logit representing the log-odds of a connection between nodes, i and j
- α : the intercept; the baseline odds of a connection between nodes i and j
- $|z_i - z_j|$: the latent distance between nodes i and j in some arbitrary distance metric^{1,2}

¹The distance metric *should* satisfy the triangle inequality. I will violate this.

²You can also see that I dropped $\beta' x_{i,j}$ as this just represents the covariates and we don't do that.

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Monopartite

$$\eta_{ij} = \alpha - |z_i - z_j| \quad (1)$$

so:

- The relation log-odds of a connection forming depends on the distance between two entities in the latent space and some baseline likelihood of a connection forming
- Note, Hoff et al. (2002) note that any arbitrary distance may be used.

Node-Specific Effects

- One may imagine that α is a node-specific parameter; thus, we'd need to redefine it as $\alpha = a_i + a_j$ in our equation resulting in:

$$\eta_{ij} = a_i + a_j - |z_i - z_j| \quad (2)$$

where

- a_x represents the baseline tendency for the x^{th} individual to connect

LSSN

Bipartite (I Think)

- We can carry forward this same logic but with a bipartite graph where α represents persons and β represents items:

$$\eta_{ij} = \alpha_i + \beta_j - \|\delta_i - \gamma_j\|^2 \quad (3)$$

where

- α_i : represents the propensity of the i^{th} individual to endorse any j -item
- β_j : represents the propensity of the j^{th} item to be endorsed by any i person
- $\|\delta_i - \gamma_j\|^2$: Represents the squared Euclidean distance between person i and item j in latent space

Why the Squared Distance?

- Let's try expanding it out:

$$\begin{aligned}\eta_{ij} &= \alpha_i + \beta_j - \|\delta_i - \gamma_j\|^2 \\ &= \alpha_i + \beta_j - \left(\|\delta_i\|^2 + \|\gamma_j\|^2 - 2\delta_i^\top \gamma_j \right) \\ &= (\alpha_i - \|\delta_i\|^2) + (\beta_j - \|\gamma_j\|^2) + 2\delta_i^\top \gamma_j\end{aligned}\tag{4}$$

set

- $\theta_i = \delta_i$ and $\mathbf{a}_j = 2\gamma_j$ then:
 - $\mathbf{a}_j^\top \theta_i = 2\delta_i^\top \gamma_j$
- $c_i = \alpha_i - \|\delta_i\|^2$ and $b_j = \beta_j - \|\gamma_j\|^2$

you get

$$\eta_{ij} = c_i + b_j + \mathbf{a}_j^\top \theta_i\tag{5}$$

$$\eta_{ij} = c_i + b_j + \mathbf{a}_j^T \boldsymbol{\theta}_i \quad (6)$$

where

- $c_i = \alpha_i - \|\delta_i\|^2$ represents the i^{th} individual's propensity to form connections after accounting for their latent position
- $b_j = \beta_j - \|\gamma_j\|^2$ represents the j^{th} item's propensity to be endorsed after accounting for its latent position
- What does this reduce to if we—for no reason in particular—assume that there is no individual propensity left over after accounting for their latent position?

$$\eta_{ij} = b_j + \mathbf{a}_j^T \boldsymbol{\theta}_i \quad (7)$$

2PL MIRT

- A 2-dimensional 2 parameter logistic (2PL) Multidimensional Item Response Theory Model (MIRT) can be formulated in a similar way:

$$\eta_{ij} = b_j + \mathbf{a}_j^T \boldsymbol{\theta}_i \quad (8)$$